

ResonatE: Unit-Norm Complex Spectra with Block-Unitary Relation Operators, and Label-Free Retrieval from the Training Graph, on `ogbl-biokg`

Cristian Malaia
`cristian.malaia@gmail.com`

Claude Fable 5
Anthropic

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<https://github.com/jinxmcg/resonate>; project page: <https://resonate.page>.*

Abstract

We report three results on `ogbl-biokg`, the Open Graph Benchmark’s biomedical link-prediction task, each as the mean and unbiased standard deviation of ten independent runs with one test evaluation per run under the official Evaluator. (A) A 27.1M-parameter embedding in which entities are complex vectors of 144 coefficients and each of the 102 (relation, direction) operators is a set of 36 independent 4×4 complex blocks initialised unitary reaches test MRR 0.8118 ± 0.0013 , above ComplEx, DistMult and RotatE at 188M parameters. (B) Four label-free retrieval features computed from the training graph alone—the similarity, under the model’s own embeddings or under raw edge overlap, between the query entity and the training entities that already hold the candidate edge—blended with the model by a per-relation weight selected on validation, reach 0.8425 ± 0.0007 . (C) A fresh 27.1M model distilled at training time from the ten models of (A), with its own features, reaches 0.8505 ± 0.0004 , which would place fifth of twelve on the public leaderboard as of September 2026 and fourth among single models, at the smallest parameter count on it. The recipe was found by a pre-registered screen-then-confirm search on the validation split in which fourteen of sixteen candidate levers were rejected, several because the architecture already supplies what the lever would add; we report them. Code, logs, frozen blend weights and checkpoints are released; the per-run test receipts reproduce deterministically from the released caches.

1 Introduction

Knowledge-graph embedding models score a triple (h, r, t) by a function of learned vectors, and on `ogbl-biokg` [9] the classic bilinear and rotational families (DistMult, ComplEx, RotatE) sit near test MRR 0.80–0.81 at roughly 188M parameters, while the current leaders reach 0.85–0.86 through learned scoring functions, auxiliary objectives or relation-aware ensembles [18]. This report describes a model with a different inductive bias and a different budget—27.1M parameters—and two label-free additions that lift it from 0.812 to 0.851 without adding a single parameter fitted to anything but the training triples.

The model, ResonatE, was designed for multi-hop reasoning: an entity is a unit-norm complex vector, a relation is a linear operator on that space, a hop is “apply the operator and renormalise”, and a chain of hops is a chain of operators. `ogbl-biokg` is a single-edge task, so here only the one-hop score matters, but the design leaves marks on the results: unit-norm renormalisation after every hop makes the usual N3 regulariser redundant, type-grouped batches make a relation-prediction auxiliary loss redundant, and exact operator composition makes path training redundant—each verified negatively in Section 6.3.

The two additions are (i) *retrieval features*: for a candidate answer c to a query $(h, r, ?)$, look up the training entities that already hold $(\cdot, r, c)$ and score c by how similar they are to h , either in the model’s own embedding space or by raw edge overlap (Section 3); and (ii) *distillation at training time*: a fresh model whose loss includes the KL divergence to the mean prediction of ten frozen models on each batch’s own candidate set (Section 4). Neither reads validation or test edges. The blend weights of (i) are selected on validation from a fixed list of at most nine candidate vectors per relation and direction; the selection is reported alongside its held-out estimate, which is within 0.001 of test on every one of the twenty runs.

Our contributions are the three ten-run results, the recipe search with its negative results, the retrieval-feature construction, and a fully scripted reproduction path with released checkpoints. The operator family itself is not a contribution; the next paragraph places it.

Relation to prior operator families. Representing a relation as a rotation or a small orthogonal block, so that mixed-relation chains compose non-commutatively, is an established line: RotatE [20] uses 1×1 complex rotations (commuting), QuatE [26] and DensE [16] quaternion rotations, Rotate3D [5] rotations in $\mathbb{R}^3$, OrthogonalE [29] block-diagonal orthogonal matrices with Riemannian optimisation and matrix-valued entities, and Knowledgebra [23] free matrix semigroups. ResonatE’s operator is nearest to OrthogonalE: block-diagonal with small complex blocks, unitary at initialisation—but the constraint is dropped afterwards (the blocks are free), the query side is renormalised after every hop while the target side keeps its norm as a popularity channel, and no path or composition loss is used (Section 6.3). We make no claim that this operator is new or better than those families; on this benchmark the block size moves validation MRR by +0.0055 ($2 \times 2 \rightarrow 4 \times 4$, 50k steps) and 8×8 is flat against 4×4 at the screen (0.7904 vs 0.7893, Section 6.2), which is the size of the effect the operator form has here. The project’s name and the “mode” vocabulary in the code are a coordinate label: the score is invariant under any block-diagonal unitary change of basis ($E \rightarrow UE$, $H_r \rightarrow UH_rU^\dagger$), no Fourier transform is taken anywhere, and nothing in the results depends on the coordinates being spectral. Nor is “complex” a claim. The one testable residue of the complex form is its parameter sharing: a complex 4×4 block on $\mathbb{C}^{144}$ and a free real 4×4 block on $\mathbb{R}^{288}$ both have 1,152 real coefficients per relation and the same 27,124,129 parameters in total. Pre-registered and run on the 12.5k-step validation screen with two seeds each, the real model scores 0.7923 against the complex model’s 0.7897, and a real 8×8 block (which contains the complex 4×4 as a special case) 0.7898; the difference is below the 0.005 we had fixed as the smallest meaningful gap and its sign is against the complex form. The complex arithmetic is a way of writing a real block model on $\mathbb{R}^{2M}$, and the released code accepts `real=True` to write it the other way.

2 Model

Entities and relations. Let N be the number of entities and $M = k^2$ with $k = 12$, so $M = 144$. The entity table is $E \in \mathbb{C}^{N \times M}$. A query entity is read through $\text{cnorm}(E_h) = E_h / \|E_h\|$; a target entity is read as the raw row E_t , so target norms are free and act as a learned per-entity popularity channel. For each of the $2R$ (relation, direction) pairs ($R = 51$, both directions trained as separate operators, $2R = 102$) the operator H_r is block-diagonal with M/b independent $b \times b$ complex blocks, $b = 4$, initialised as random unitaries by QR decomposition and then trained freely. A hop is

$$z = \text{cnorm}(H_r \text{cnorm}(E_h)), \quad (1)$$

and the score of candidate t is

$$s(h, r, t) = \text{Re}\langle z, E_t \rangle \cdot e^\tau, \quad (2)$$

with a single learned log-temperature τ . Blocks larger than 1×1 do not commute, so mixed-relation chains are representable; $b = 4$ was chosen on validation (Section 6.2).

Parameter count. $N \cdot M$ complex entity coefficients ($93,773 \times 144$), $2R \cdot (M/b) \cdot b^2$ complex operator coefficients ($102 \times 36 \times 16$) and τ : counting each complex coefficient as two reals, as the leaderboard does for ComplEx, 27,124,129 parameters. The literal number of parameter elements is 13,562,065.

Training. Entity ids are per type in OGB; we flatten them with per-type offsets so that each relation’s head and tail types map to contiguous ranges. A training batch is 2,048 triples of a single relation (relations sampled in proportion to their frequency), scored in one random direction against a single shared set of 4,096 negatives drawn uniformly from the target type’s range—the same candidate distribution the Evaluator uses. The loss is the sampled softmax cross-entropy over [positive | negatives] plus a trajectory term $0.1 \|z - E_t\|^2$ on the hopped state, with gradient-norm clipping at 1. Adam [12] at learning rate $5 \cdot 10^{-3}$ with cosine decay [15] for 50,000 steps (≈ 21 epochs). Training takes about four minutes on an RTX 5090 and eighteen on a GTX 1080 Ti.

3 Retrieval features from the training graph

The Evaluator scores each query $(h, r, ?)$ over 501 candidates: the true tail and 500 fixed, type-matched negatives (and symmetrically for heads). For every candidate c we form features from the *holders* of c under r in the training graph, $\mathcal{H}_r(c) = \{h' : (h', r, c) \in \mathcal{T}_{\text{train}}\}$, and the similarity between the query entity h and those holders:

$$\text{analogy}(h, r, c) = \max_{h' \in \mathcal{H}_r(c)} \cos(\text{cnorm}(E_h), \text{cnorm}(E_{h'}))^3, \quad (3)$$

$$\text{jaccard}(h, r, c) = \max_{h' \in \mathcal{H}_r(c)} \frac{|\mathcal{N}(h) \cap \mathcal{N}(h')|}{|\mathcal{N}(h) \cup \mathcal{N}(h')|}, \quad (4)$$

where $\mathcal{N}(\cdot)$ is the typed training-edge set of an entity, plus the two “top-3” variants that replace the max by the mean of the three largest similarities. The cube on the cosine sharpens the analogy score; it is the only shape choice and was made on validation. A candidate with no holders gets the minimum score. Both families read training edges only—validation and test edges are never looked up—so they are as label-free as the embedding itself; the Jaccard pair does not use the model at all. This is case-based reasoning in the model’s own geometry (analogy) and in the raw graph (Jaccard): the embedding compresses an entity’s neighbourhood at training time, the graph remembers it exactly at query time. Alone, on validation, the four features reach MRR 0.645 (analogy), 0.751 (analogy top-3), 0.671 (Jaccard) and 0.764 (Jaccard top-3).

Blend. For each member (the model plus the four features) the 501 candidate scores of a query are z -normalised. The blended score is a weighted sum with one weight vector $w \in \Delta^4$ per (relation, direction) group. The candidate set for a group is fixed: uniform, the uniform mean of the top-1, -3, -5 or -9 members by that group’s validation MRR, and $\text{softmax}(\eta \cdot \text{MRR})$ for $\eta \in \{20, 50, 100\}$; the choice is made hierarchically—a global vector first, then one per relation family (the `drug-drug_*` and `protein-protein_*` relations pooled), then one per relation, each level adding the parent’s choice to its candidate list and a group only receiving its own weights if it has at least 4,000 validation rows. This is a selection among at most nine vectors per group, not a gradient fit. Its optimism is measured by the same procedure with the weights fit on a random half of the validation triples (a triple’s two directions stay together) and evaluated on the other half (`ensemble_weights.py`); the committed test shot refits on the full validation split and applies the frozen weights once (`freeze_test.py`). The frozen weights are released with each receipt.

4 Distillation at training time

Ten checkpoints are a deployment cost, and a uniform score average of the ten row-A models reaches only 0.8428 (one test shot, retired). Instead we train a fresh model of the same architecture and recipe whose loss adds, on every batch,

$$\lambda T^2 \text{KL}(\text{softmax}(\bar{\ell}/T) \parallel \text{softmax}(\ell/T)), \quad (5)$$

where ℓ are the student’s logits over the batch’s [positive | 4,096 negatives] and $\bar{\ell}$ the mean of the ten frozen teachers’ logits on the same candidates [8], with $\lambda = 1$, $T = 1$. The teachers see only the training triples the student sees; the student runs alone at inference. Ten students (seeds 0–9, the same ten teachers) reach validation MRR 0.8287 ± 0.0006 alone—+0.016 over their teachers—and 0.8505 ± 0.0004 on test with their own retrieval features.

5 Protocol

Data and metric. `ogbl-biokg`: 93,773 entities of five types (disease 10,687, drug 10,533, function 45,085, protein 17,499, side effect 9,969), 51 relations, 4,762,678 training, 162,886 validation and 162,870 test triples. Filtered MRR over 500 type-matched negatives per direction via `ogb.linkproppred.Evaluator` (`ogb` 1.3.6). Our loader asserts that the three splits are pairwise disjoint as loaded (`verify.py`).

Pre-registration and test reads. Every configuration’s expected test range was written in the project log before its first test read. All comparisons between recipes, features, blend rules and distillation settings were made on validation. The test split is read once per run: row A by the trainer’s final evaluation, rows B and C by the committed blend shot; test score caches are written without computing any per-member test metric. Nine test reads were made during development outside the three rows, each a single shot of a configuration chosen on validation and none re-run: the 25k-step baseline (0.8024); 50k steps with 2×2 blocks, seeds 0 and 1 (0.8084, 0.8080); the final recipe’s seeds 0 and 1 trained on the 1080 Ti (0.8134, 0.8123; their 5090 retrains are row A seeds 0 and 1, 0.8135 and 0.8130); the uniform ten-model ensemble (0.8428); seed 0 with three analogy features (0.8368); a 48M single model ($k = 16$, 16,384 negatives, per-entity bias) with its features (0.8472); and the same 48M recipe distilled from twelve checkpoints, with features (0.8535). The last two are exploratory and are not part of any reported row.

Statistics. Mean $\pm$ unbiased standard deviation (`torch.std`) over seeds 0–9. Row C’s ten students share the same ten teachers, so its deviation reflects student-seed variance only.

Hardware. Rows A–C were trained on rented RTX 5090 (32 GB) instances with `torch` 2.11.0 (CUDA 12.8); the retrieval features of seeds 7–9 of row B and all verification ran on a GTX 1080 Ti (11 GB) with `torch` 2.6.0 (CUDA 12.4). Checkpoints trained on one machine re-evaluate on the other to within $3 \cdot 10^{-4}$ MRR. The analogy features need about 6 GB of RAM and 25–40 minutes per split per seed on one CPU core.

6 Results

6.1 Main result

Table 1 gives the three rows. The three clusters of ten do not overlap: each step (+0.031, +0.008) is an order of magnitude larger than the seed noise. The held-out estimate of the blend is within 0.001 of test on every run (mean gap +0.0004 for B, +0.0007 for C), so the selection on validation carries no measurable optimism to test.

Table 1: `ogbl-biokg`, official Evaluator, ten runs per row, one test read per run. “valid” for rows B and C is the in-sample value of the blend weights, which are selected on the full validation split; “held-out” is the same procedure fit on half of validation and measured on the other half. Every row has 27,124,129 parameters.

	test MRR	valid MRR	held-out	hits@1	hits@3	hits@10
A. single model	0.8118 ± 0.0013	0.8124 ± 0.0013	—	0.741	0.862	0.939
B. A + retrieval features	0.8425 ± 0.0007	0.8434 ± 0.0007	0.8429 ± 0.0007	0.783	0.884	0.951
C. distilled + retrieval	0.8505 ± 0.0004	0.8515 ± 0.0004	0.8512 ± 0.0005	0.794	0.890	0.952

Table 2: Public leaderboard for `ogbl-biokg` as fetched on 2026-09-02 [18], with published parameter counts, and where the three rows would sit.

	method	test MRR	params
1	RelEns [25] (ensemble)	0.9618	849M
2	ComplEx ²	0.8583	188M
3	UniBi	0.8550	182M
4	AutoBLM [28]	0.8536	192M
	ResonatE C: distilled 27M + retrieval	0.8505	27M
5	ComplEx-RP [3]	0.8492	188M
	ResonatE B: 27M + retrieval	0.8425	27M
6	TripleRE [24]	0.8348	470M
7	AutoSF [27]	0.8309	94M
8	PairRE [2]	0.8164	188M
	ResonatE A: 27M single model	0.8118	27M
9	ComplEx [21]	0.8095	188M
10	DistMult [22]	0.8043	188M
11	RotatE [20]	0.7989	188M
12	TransE [1]	0.7452	188M

Table 2 places the rows on the public board. Row C would be fifth overall and fourth among single models; every ResonatE row is the smallest model on the board by $3.5\times$ (AutoSF is 94M; every other entry is 182M or more).

6.2 How the recipe was found

The starting point was a hand-picked recipe from smaller graphs: $k = 12$, 2×2 blocks, 4,096 negatives, learning rate $5 \cdot 10^{-3}$ cosine, 25,000 steps. Each default was then treated as a hypothesis under a tiered protocol written down before each run: a 12,500-step *screen* on validation, promotion only above the current recipe by $+0.004$ MRR, then a 50,000-step *confirmation*. Sixteen levers went through it; two survived.

Budget. Validation MRR at 6,250 / 12,500 / 25,000 / 50,000 steps (each with its own cosine schedule) is 0.7351 / 0.7829 / 0.8032 / 0.8085: the inherited budget had not plateaued. The gain per doubling shrinks fast enough that 100k projects at $\leq +0.002$, so the budget was doubled once and frozen.

Capacity: blocks, not embedding width. Capacity can go into k (every entity’s embedding grows; millions of parameters) or into the relation blocks ($2 \times 2 \rightarrow 4 \times 4$; 118k real parameters in total). In 2,000-step probes validation MRR grows log-linearly in k with no plateau (Table 3), but the gain compresses with budget: $k = 16$ ’s $+0.025$ at 2k steps is $+0.003$ at 50k, and even $k = 32$ (192M parameters) misses the promotion bar. Blocks are the opposite: their $+0.0064$ screen gain barely compresses to $+0.0055$ at 50k (0.8140)—expressivity rather than capacity, at

Table 3: Capacity sweep on validation, seed 0, 2×2 blocks except the last row. Parameters count complex coefficients as two reals.

config	params	2k-step probe	50k steps
$k = 4$	3.0M	0.3722	—
$k = 8$	12.0M	0.5350	—
$k = 12$	27.1M	0.5869	0.8085
$k = 16$	48.1M	0.6115	0.8118
$k = 20$	75.2M	0.6267	—
$k = 24$	108.3M	0.6349	—
$k = 32$	192.5M	0.6510	0.8129
$k = 12$, 4×4 blocks	27.1M	—	0.8140

Table 4: Levers rejected on validation. Screen: 12,500 steps on the 4×4 recipe (reference 0.7893); confirm: 50,000 steps (reference 0.8140). “✓” marks a passed screen.

lever	screen	confirm	verdict
relation-prediction aux loss, $\lambda = 0.25/1/4$	0.7886/0.7849/0.7741	—	monotone negative; type-grouped batches already separate relations
N3 regularisation, $w = 10^{-3}/10^{-2}/10^{-1}$	0.7902/0.7912/0.7894	—	real interior peak, half the promotion bar
compositional (path) training, $p = 0.25/0.5$	0.7711/0.7475	—	hops compose exactly already; chains tax the update budget
16,384 negatives	0.7963 ✓	0.8148 (+0.0008)	convergence-speed effect
tied reverse operators (adjoint)	0.7936 ✓	0.8128 (−0.0012)	reverse blocks are not slack
one-cycle schedule, max lr $3 \cdot 10^{-2}$	0.7474 [†]	—	update count is the bottleneck
uniform 8×8 blocks	0.7904	—	flat; 4×4 is the knee
unitary (phase-only) relations	< 90% at every depth (synthetic)	—	off-circle moduli are load-bearing
50k budget (adopted)	—	0.8085 vs 0.8032 (25k)	
4×4 blocks (adopted)	0.7893 ✓	0.8140 ✓	survives the budget

[†]screened on the 2×2 baseline against its own cosine 0.7829.

roughly 1/300 the parameter price per MRR point. Stacking $k = 16$ on 4×4 blocks adds almost nothing; the two knobs buy overlapping capacity. Uniform 8×8 blocks are flat (0.7904 at the screen vs 0.7893).

6.3 What did not work

The remaining fourteen levers—mostly the tricks other leaderboard entries rely on—went through the same screen-then-confirm pipeline on top of the adopted recipe (Table 4; the 12,500-step reference is that recipe’s own 0.7893). Two patterns account for them. First, *the architecture pre-empts the trick*: a relation-prediction auxiliary loss [3] is monotonically harmful because type-grouped batches already force the model to separate relations; N3 regularisation [13] has a real but tiny interior optimum because renormalising after every hop already disciplines magnitudes; compositional path training [6] has nothing to repair because hops compose exactly by construction and the chains only spend the update budget. Second, *screen-tier wins vanish at the plateau*: more negatives and tied reverse operators both pass the 12,500-step screen and are flat or negative at 50k—they speed convergence without raising the ceiling. A one-cycle schedule loses to cosine, and phase-only (unitary) relation operators never reach 90% accuracy at any depth in the synthetic chain testbed: the moduli off the unit circle are load-bearing.

Table 5: Development shots (seed 0 unless noted; validation is the held-out estimate of the blend), and the two ten-run rows built from them.

configuration	valid	test
single model, $k = 12$, 27M	0.8135	0.8118 (10-seed mean)
+ 3 analogy features (shot 1)	0.8370	0.8368
+ Jaccard pair (analogy from three seeds’ embeddings)	0.8441	—
27M + its own analogy, top-3 and Jaccard pair (shot 3)	0.8434	0.8433
row B: the same, ten seeds	0.8429 ± 0.0007	0.8425 ± 0.0007
48M single ($k = 16$, 16k neg., entity bias) + own features (shot 2)	0.8474	0.8472
48M distilled from 12 checkpoints + own features (shot 4)	0.8540	0.8535
27M distilled from the ten row-A models + own features (shot 5)	0.8516	0.8509
row C: the same, ten student seeds	0.8512 ± 0.0005	0.8505 ± 0.0004
uniform ten-model score average (retired)	—	0.8428

6.4 Retrieval and distillation, step by step

Table 5 lists the single-seed shots that led to rows B and C. Each configuration was chosen on validation and read on test once. Three analogy features on top of seed 0 give 0.8368; adding the Jaccard pair lifts the held-out estimate to 0.8441 (with analogy features that borrowed three seeds’ embeddings); a model with its *own* four features gives 0.8433—above the uniform ten-model ensemble’s 0.8428 with one model. The two rows that go on the board are these last two recipes re-run as ten-seed protocols.

Two levers rejected in Section 6.3 returned once the recipe was stronger: 16,384 negatives and $k = 16$ are each flat alone but reach 0.8171 validation together, and a per-entity score bias adds +0.003 on top; that 48M model with its features gives 0.8472, and distilled from twelve checkpoints 0.8535. We did not run them as ten-seed rows. What did not help, all measured held-out before any shot: $k = 24$ (parity at twice the parameters), a confidence gate on the blend weights (−0.0015), cross-relation neighbourhood overlap (+0.0002), compositional training (−0.005).

Where the errors are. On the rarest relations the lookup beats the embedding outright: on side effects with at most 19 training edges the analogy feature alone reaches held-out MRR 0.0175 against the 27M model’s 0.0147 (blend 0.0196; paired bootstrap significant). Of the remaining errors 36% are drug→side-effect tails, where even a 22-member ensemble scores 0.17—a floor set by the data. Across the ten seeds the truth-versus-runner-up margin has a constant spread of 0.28 and only its mean moves: seeds disagree on which near-tied candidate wins, not on what they know, which is why more seeds were never going to close the gap and why the deployable unit is one table plus a lookup.

7 Beyond the single-edge task

The leaderboard credential is a one-hop score; the architecture was built for chains, and the same code trained on graphs with multi-hop questions shows the properties the benchmark cannot. We summarise them briefly; they are not part of the submission.

Depth. On synthetic worlds with exact ground truth at every depth, models trained only on chains of depth ≤ 4 ($k = 8$, 25k steps, three seeds, 2,000 chains per depth) are read out to depth 48. Three results, in decreasing order of what they show. *One relation applied n times*, on 1,000 entities with seven random permutations as relations: the diagonal model holds 0.987 at depth 8 and crosses 90% between depth 10 and 12 and 50% between 16 and 20; the 4×4 -block model 0.963 at depth 8. This is a statement about powers of a single operator, not about composition.

A different relation at every hop, so that order matters, on the same random permutations: neither the block model nor a dense 64×64 operator learns depth 2 (0.21 and 0.65 accuracy, ≈ 0 beyond), because random permutations share no low-dimensional structure; the block model does not compose arbitrary relations. *A different relation at every hop on relations that form a group*: a turtle on a 16×16 torus with forward, back, turn left and right, strafe left and right, and U-turn, i.e. 1,024 (heading, x , y) states and rigid motions generating $\mathbb{Z}_{16}^2 \rtimes \mathbb{Z}_4$, whose irreducible representations are at most four-dimensional. Half of all depth-2 chains change their answer when reordered, and a commuting diagonal model reaches 0.75 at depth 2. The 4×4 -block model trained on depth ≤ 4 reaches 0.949 at depth 8, crosses 90% between depth 10 and 12 and 50% between 20 and 24 (seed s.d. ≤ 0.003 to depth 8); the dense operator reaches further (0.995 at depth 8, 0.885 at depth 20), so the block size is a ceiling rather than the mechanism. The allowed statement is therefore that the model composes non-commuting relations beyond its trained depth when the relations form a group with a low-dimensional representation, and not for arbitrary permutations; whether the relations of a real graph are closer to the former or the latter is not measured. Both mixed-relation experiments were pre-registered after a first external review asked for them, and the earlier single-relation number was the only one we had quoted. Mapping natural-language questions onto chains of relation operators is outside this report.

Latency and a compiled index. Because the per-hop renormalisation is a scalar, a metapath’s operator chain collapses into one block-diagonal operator (2,304 bytes for the 2×2 models), and the readout is a real inner product—maximum-inner-product search over a real $N \times 2M$ table. On DRKG [10] (97,238 nodes, 5.87M edges, a 28.1M-parameter model) batched multi-hop queries take 0.08–0.11 ms at every depth 1–6 while exact traversal grows with fan-out, up to $45\times$ slower at depth 5–6; compiled scores equal hop-by-hop scores to 10^{-5} with identical top-10 sets over 1,867 queries, and an HNSW index [11, 17] over the table gives recall@10 0.993 against the exact readout. Storing the table in fp16 (848 bytes per row including HNSW links) leaves the top-10 identical on all 3,334 queries over two graphs; int8 costs a little (top-1 agreement 0.98), int4 breaks retrieval (0.70).

Oracle validation. On Hetionet [7] and DRKG every top answer is checked against an exact graph oracle and labelled as a citable path or as a novel hypothesis; on the trained metapath libraries the top answer is oracle-verified 94.6–100% of the time across all ten DRKG paths and 99–100% on all eight Hetionet paths. Fine-tuning 496 held-out *treats* edges into a trained DRKG model took 3.7 GPU-minutes, absorbed all of them to rank 1 (MRR 0.147 $\rightarrow$ 1.000) with a worst metapath regression of -0.008 .

Scaling. A training step touches about 10k rows whatever N is; what kept the table resident was dense Adam state. Replacing it for the entity table by row-wise Adagrad [4, 14] (one accumulator per row, no decay, untouched rows left bit-identical) costs nothing measurable on Hetionet (hits@10 0.782 vs 0.782 over three seeds) and lies within one standard error on DRKG (link-prediction MRR 0.157 vs 0.163 on 496 edges). From measured per-row costs, 100M entities at $k = 12$ would need a 115 GB fp32 training table, 0.4 GB of optimiser state, an 85 GB fp16 HNSW serving index (or about 5 GB of RAM plus NVMe in a DiskANN layout [19]) and roughly 7 GPU-hours for 20 epochs of 500M edges on one 5090-class card; the remaining work is sparse row-gradient plumbing and type-sharding, and none of it has been run.

8 Limitations and disclosures

The blend weights of rows B and C are selected on the full validation split; we report the in-sample validation number and the held-out estimate of the same selection, which is within 0.001

of test on all twenty runs. Row C’s ten students share one teacher pool. The retrieval features cost 25–40 CPU-minutes per split per seed and roughly 6 GB of RAM, and at inference a query needs the holder lists of its 501 candidates—cheap, but a graph lookup the pure embedding does not need. Nine development test reads are listed in Section 5. Training is seeded but CUDA kernels are not bit-deterministic, so a retrained checkpoint reproduces its MRR to seed-level noise rather than bit-for-bit; the blend stage is exact. No external data was used.

9 Reproducibility

The repository <https://github.com/jinxmkg/resonate> contains the model and training code, the feature and blend scripts, the per-run logs and receipts behind Table 1, the frozen blend weights, three scripts that reproduce rows A–C end to end, a `verify.py` that asserts split disjointness and re-evaluates the released checkpoints against their logs, and a release with all twenty checkpoints. `summarize.py` recomputes every statistic in this report from the receipts.

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